Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 19

Understand logistic regression for a given dataset

Experiment 19: Understand logistic regression for a given dataset

Aim

To apply logistic regression to the given dataset.

Source Code

```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df
df.head()
features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
features
label
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
for i in range(1,401):
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.
model=LogisticRegression()
model.fit(x_train,y_train)
train_score=model.score(x_train,y_train)
test_score=model.score(x_test,y_test)
if test_score>train_score:
print("Test {} Train{} Random State {}".format(test_score,train_score,i)
Test 0.6875 Train0.63125 Random State 3
Test 0.7375 Train0.61875 Random State 4
Test 0.6625 Train0.6375 Random State 5
Test 0.65 Train0.640625 Random State 6
Test 0.675 Train0.634375 Random State 7
Test 0.675 Train0.634375 Random State 8
Test 0.65 Train0.640625 Random State 10
Test 0.6625 Train0.6375 Random State 11
Test 0.7125 Train0.625 Random State 13
Test 0.675 Train0.634375 Random State 16
Test 0.7 Train0.628125 Random State 17
Test 0.7 Train0.628125 Random State 21
Test 0.65 Train0.640625 Random State 24
Test 0.6625 Train0.6375 Random State 25
Test 0.75 Train0.615625 Random State 26
Test 0.675 Train0.634375 Random State 27
Test 0.7 Train0.628125 Random State 28
Test 0.6875 Train0.63125 Random State 29
Test 0.6875 Train0.63125 Random State 31
T t 0 6625 T i 0 6375 R d St t 37
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)
print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))
from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

Output - Experiment 19

User ID Gender Age EstimatedSalary Purchased
0 15624510 Male 19 19000 0
1 15810944 Male 35 20000 0
2 15668575 Female 26 43000 0
3 15603246 Female 27 57000 0
4 15804002 Male 19 76000 0
... ...
395 15691863 Female 46 41000 1
396 15706071 Male 51 23000 1
397 15654296 Female 50 20000 1
398 15755018 Male 36 33000 0
399 15594041 Female 49 36000 1
400 rows × 5 columns
User ID Gender Age EstimatedSalary Purchased
0 15624510 Male 19 19000 0
1 15810944 Male 35 20000 0
2 15668575 Female 26 43000 0
3 15603246 Female 27 57000 0
4 15804002 Male 19 76000 0
array([[19, 19000],
[35, 20000],
[26, 43000],
[27, 57000],
[19, 76000],
[27, 58000],
[27, 84000],
[32, 150000],
[25, 33000],
[35, 65000],
[26, 80000],
[26, 52000],
[20, 86000],
[32, 18000],
[18, 82000],
[29, 80000],
[47, 25000],
[45, 26000],
[46, 28000],
[48 29000]
array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
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Output - Experiment 19 (continued)

1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1,
0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1,
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0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0,
1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1,
0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1,
1, 1, 0, 1], dtype=int64)
LogisticRegression()
0.834375
0.9125
precision recall f1-score support
0 0.85 0.93 0.89 257
1 0.84 0.71 0.77 143
accuracy 0.85 400
macro avg 0.85 0.82 0.83 400
weighted avg 0.85 0.85 0.85 400

Result - Experiment 19

Logistic regression was applied to the given dataset.